

Current Trends in Ultrasound Wearables: Spotlight on System Architecture

David Weik , *Member, IEEE*, Richard Nauber , *Senior Member, IEEE*, Erik Kaiser ,
Nele Kirsch , *Graduate Student Member, IEEE*, Robert Kunz , Lennart Schierling ,
Christoph Leitner , *Member, IEEE*, Luca Benini , *Fellow, IEEE*, Hsiao-Chuan Liu , *Member, IEEE*,
Qifa Zhou , *Fellow, IEEE*, Jochen Hampe , Gerhard Fettweis , Moritz Herzog , *Member, IEEE*,
and Christian Kupsch , *Member, IEEE*

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Abstract—Wearable ultrasound sensing systems are rapidly emerging for precise, continuous, and intuitive biomedical monitoring and human-in-the-loop interaction in healthcare, industry, and rehabilitation. These systems must operate under stringent constraints on size, weight, and power while delivering actionable physiological and functional information. Advances in micromachined transducers, conformable electronics, low-power signal processing, and edge artificial intelligence (AI) have enabled the first generation of wearable prototypes, yet integration

of hardware and software at the system-level remains a major barrier to mass deployment. This review maps the technology readiness and architectures of wearable ultrasound systems, and examines critical design trade-offs, including edge versus cloud-based processing and pulse-echo versus coded signal approaches. We identify recurring design principles and argue that modular, scalable, and reusable platforms are key to lowering development barriers and accelerating translation from prototypes to commercial deployment across healthcare, industrial, and consumer domains.

Index Terms—Wearable ultrasound, wearable health monitoring, human-machine interfaces, edge computing, coded signals, wireless systems, system design.

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David Weik, Robert Kunz, Lennart Schierling, and Christian Kupsch are with Measurement, Sensor and Embedded Systems Laboratory (MSE Lab), TU Bergakademie Freiberg, 09599 Freiberg, Germany (e-mail: christian.kupsch@mse.tu-freiberg.de).

Richard Nauber, Nele Kirsch, and Gerhard Fettweis are with the Vodafone Chair Mobile Communications Systems, Department of Electrical Engineering, TU Dresden, 01069 Dresden, Germany.

Erik Kaiser and Moritz Herzog are with the Else Kröner Fresenius Center for Digital Health, TU Dresden Faculty of Medicine Carl Gustav Carus, 01307 Dresden, Germany.

Christoph Leitner is with the Department of Information Technology and Electrical Engineering, ETH Zurich, 8092 Zurich, Switzerland.

Luca Benini is with the Department of Information Technology and Electrical Engineering, ETH Zurich, 8092 Zurich, Switzerland, and also with the Department of Electrical, Electronic, and Information Engineering, University of Bologna, 40126 Bologna, Italy.

Hsiao-Chuan Liu is with the Department of Biomedical Engineering, University of Massachusetts Amherst, Amherst, MA 01002 USA.

Qifa Zhou is with the Department of Ophthalmology and Biomedical Engineering, Viterbi School of Engineering, University of Southern California, Los Angeles, CA 90007 USA.

Jochen Hampe is with the Medical Department 1, University Hospital Dresden, TU Dresden Faculty of Medicine Carl Gustav Carus, 01307 Dresden, Germany.

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I. INTRODUCTION

WEARABLE ultrasound is emerging as a key technology to meet the growing demand for continuous, unobtrusive physiological monitoring and human-in-the-loop interaction for healthcare, industry, and rehabilitation [1], [2], [3]. In these domains, common drivers include aging populations, the shift towards personalized and preventive medicine, and the increasing integration of human sensing into adaptive technical systems. In healthcare, this includes longitudinal, non-invasive monitoring of physiological parameters such as blood pressure or tissue perfusion in both clinical and home settings [4]. In industry, human-in-the-loop interaction integrates human action and decision-making into robotic and augmented-reality systems [5], while rehabilitation and personal training increasingly rely on home-based applications with real-time tracking and adaptive feedback [6]. Together, these trends require sensing technologies that operate reliably over extended periods, adapt to a wide range of real-world conditions, and provide meaningful physiological insight in both supervised and unsupervised environments.

Among wearable sensing modalities, ultrasound stands out by combining diagnostic depth, tissue specificity, and safety with the potential for continuous operation. While a range of alternative modalities is actively explored [7], [8], [9], [10], few offer comparable access to hemodynamic and structural information while remaining compatible with non-invasive, body-conformal deployment. Significantly, the realization of wearable

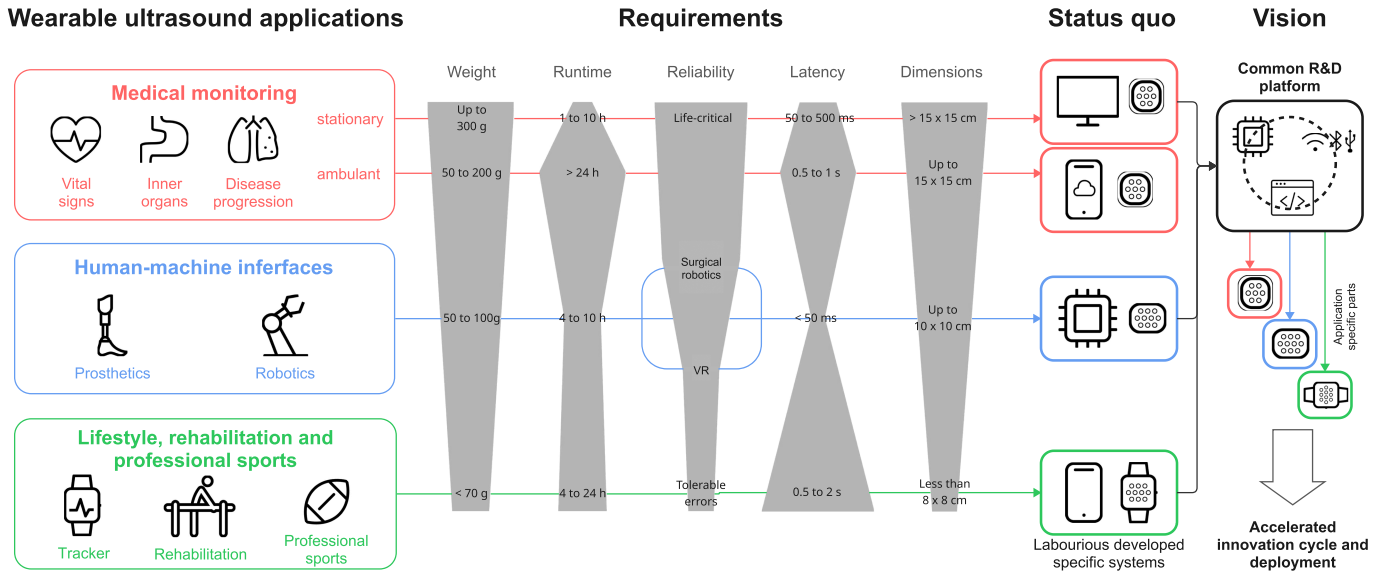


Fig. 1. Overview of wearable ultrasound systems across application domains and their requirements. Present R&D emphasizes application-specific systems, while shared platforms (vision) based on common design rules and directions could accelerate early-stage development but remain unavailable or at a very early stage.

ultrasound systems has only recently become feasible, enabled by rapid advances in supporting technologies:

- Novel micromachined ultrasound transducers now offer low-cost, wafer-scale fabrication [11], [12], [13],
- Flexible, conformable electronic substrates now allow ultrasound sensors [14] and circuitry [15] to adapt to the body's surface [16],
- Highly integrated and energy-efficient signal processing architectures support continuous operation within the strict power constraints of wearable systems [17], [18], [19], [20], and
- Embedded artificial intelligence processors and algorithms allow on-device inference and real-time decision-making [21], [22].

Together, these advances establish the technological foundation for wearable ultrasound applications, (see Fig. 1).

Wearable ultrasound systems inherit many of the core design requirements found in clinical [23], [24] and portable [25], [26], [27] scanners. These include generating and acquiring ultrasound signals across channel counts ranging from only a few single-elements to hundreds or even thousands in array systems, in combination with the need for real-time processing for signal analysis, imaging, and information extraction [28]. In addition, all ultrasound systems require bio-compatible packaging and must comply with safety regulations regarding thermal and acoustic exposure [29].

In wearable systems, these requirements are complemented by additional considerations on weight, runtime, unsupervised reliability, latency, and form factor [30], which considerably restrict the design space (see Fig. 1). Towards deployment, many reviews on wearable ultrasound have addressed advances in transducer materials, novel application domains or clinical translation [1], [2], [3], [16], [29], [31], [32], [33], [34]. System-level considerations on how to combine these advances and

meet the requirements all together have so far received limited attention. Given its importance for application development and commercialization, this raises two central questions that frame our review.

First, which design strategies best address the constraints of wearability, and what factors determine the transition of ultrasound systems from prototypes to commercial products? Transitioning wearable ultrasound systems from research prototypes to widely adopted tools in healthcare, industry, or personal use requires technical feasibility, but also economic viability. For example, the miniaturization of ultrasound hardware often depends on custom silicon integration [20], which demands substantial upfront investments, only justified if large-scale deployment is foreseeable. In this review, we examine this investment threshold by assessing the potential market size alongside the current development status of wearable ultrasound applications with their respective technology readiness levels (TRL). We aim to highlight the application domains that would benefit the most from near-term system-level innovation of ultrasound wearables and examine how common architectural and design strategies may accelerate adoption.

Second, how do current implementations address modularity, reuse, and scalability in design? Current research shows that the development of wearable ultrasound systems remains highly ad hoc: most prototypes are application-specific, with custom hardware and software tailored to a single use case [35], [36], [37]. While this allows for deep optimization, it leads to fragmented solutions, with each project requiring separate transducer interfaces, acquisition circuitry, and processing pipelines. As a result, each new project often requires building up core system components from scratch [15], [38], raising the barrier for interdisciplinary teams to pursue application-level research. In contrast, modular and reusable platforms could accelerate development across application domains, as they take development overhead

from interdisciplinary groups and enable them to focus on innovation rather than hardware integration. Such platforms could lower the barrier to entry for application-oriented researchers, including clinicians, rehabilitation scientists, or HMI developers, who wish to explore wearable ultrasound but lack expertise in hardware design. By decoupling system-level engineering from domain-specific innovation, modular platforms shorten development cycles, accelerate prototyping, and broaden participation across disciplines, as demonstrated by stationary ultrasound research platforms [39], [40], [41], [42]. Our goal is to analyze the system-level landscape of ultrasound wearables by identifying recurring architectural patterns, modular design elements, and points of variability that could converge into common platforms (see Fig. 1).

This review does not propose a specific platform design; rather, it provides a structured synthesis to inform discussions about shared components and potential convergence. It surveys developments published from 2018 to mid-2025, drawing on academic literature, patents, technical documentation, and public information on commercial products. By combining insights from research and industry, this review aims for an integrated overview of system-level architectures and their emerging applications. The scope is limited to sensing applications of wearable ultrasound and excludes therapeutic uses, due to their fundamentally different system requirements. We first analyze potential application domains in terms of market relevance and current TRL (Section II). We then outline the fundamental design principles and building blocks of ultrasound systems, emphasizing aspects particularly relevant to wearable implementations (Section III), before reviewing the state-of-the-art in key trends, implementations, and system concepts (Section IV). Finally, we synthesize the findings by linking specific applications to architectural demands and market potential, and identify cross-domain opportunities and promising directions for future research (Section V).

II. FORTHCOMING APPLICATIONS AND MARKET POTENTIAL

The following section aims to provide a comprehensive overview of technological maturity, commercial potential, and application-specific trends as a contextual groundwork to the system architecture of ultrasound wearables. Therefore, we categorized the landscape of wearable ultrasound applications into three main domains: (1) Medical monitoring, (2) human-machine interfaces (HMI), and (3) monitoring for rehabilitation, sports and lifestyle. For each sector, we analyze the market size and the compound annual growth rate (CAGR) to judge commercial potential. Specific use cases and functional areas are assessed by evaluating their current stage of development using Technology Readiness Levels (TRL) [43]. The TRL assessment primarily reflects the degree of application-specific validation, with emphasis on the extent to which the intended functionality has been demonstrated in relevant environments. Accordingly, the assessment is not limited to fully wearable solutions. Instead, even stationary solutions with the future goal of a wearable application are included and denoted by a lower

TRL. Further details of the TRL assessment are given in the supplement Section 1. An overview of the different use cases, their TRL assessment, and their commercial potential is provided in Fig. 2.

A. Medical Monitoring

The most prominent and impactful applications of wearable ultrasound are probably within the medical field, with systems being developed for continuous health monitoring, diagnostics, and disease treatment [30]. Wearable medical devices substantially support the early detection, as well as long-term monitoring of various diseases. They enable continuous and context-aware acquisition of physiological parameters, not only within, but also outside of clinical settings [31]. This extends diagnostic capabilities beyond sporadic in-clinic measurements and facilitates more informed medical decisions. In addition, such systems contribute to the relief of healthcare infrastructures, promote health awareness and patient empowerment through increased personal involvement in disease management [29]. The market for medical wearables is expected to exceed \$3.7 billion by 2030, with a Compound Annual Growth Rate (CAGR) of 8.2% [44], see Fig. 2(a). Ultrasound-based solutions are considered a significant contribution to this growth due to their non-invasive nature and high diagnostic accuracy.

In the field of continuous health monitoring, *blood pressure monitoring* plays a pivotal role [45]. In clinical practice, this parameter is essential for assessing patient status and is obtained either invasively, with associated infection risk, or non-invasively, yielding only intermittent measurements. To pursue a continuous and non-invasive blood pressure monitoring, [46] evaluated sequential ultrasound pulses in the forearm or the carotid artery with a wearable system. The method was validated in various clinical settings on human subjects, including outpatient evaluations of postural homeostasis, the cardiac catheterization laboratory, and the intensive care unit. Therefore, the technology can be classified as TRL 7. To further advance the research system used, the group also demonstrated a fully wearable device in a complementary study [15].

Blood flow velocity is also a crucial parameter for continuous cardiovascular health monitoring. Blood flow velocity describes the speed at which blood travels through the arteries, with deviations that may indicate cardiovascular disorders such as stenosis or heart failure. Ref. [47], [48] showed that wearable Doppler ultrasound can reliably measure blood flow velocity, for which [35] commercialized FloPatch for real-time hemodynamic monitoring. FloPatch has been validated in operational settings and is ready for large-scale deployment and manufacturing, corresponding to TRL 8.

Furthermore, ultrasound enables the measurement of both *respiratory rate* and *tidal volume*. Together, these parameters evaluate whether respiration provides adequate ventilation, which is particularly important after extubation and in postoperative or otherwise vulnerable patients, where reliable, noninvasive tidal-volume monitoring remains unavailable. Respiratory rate

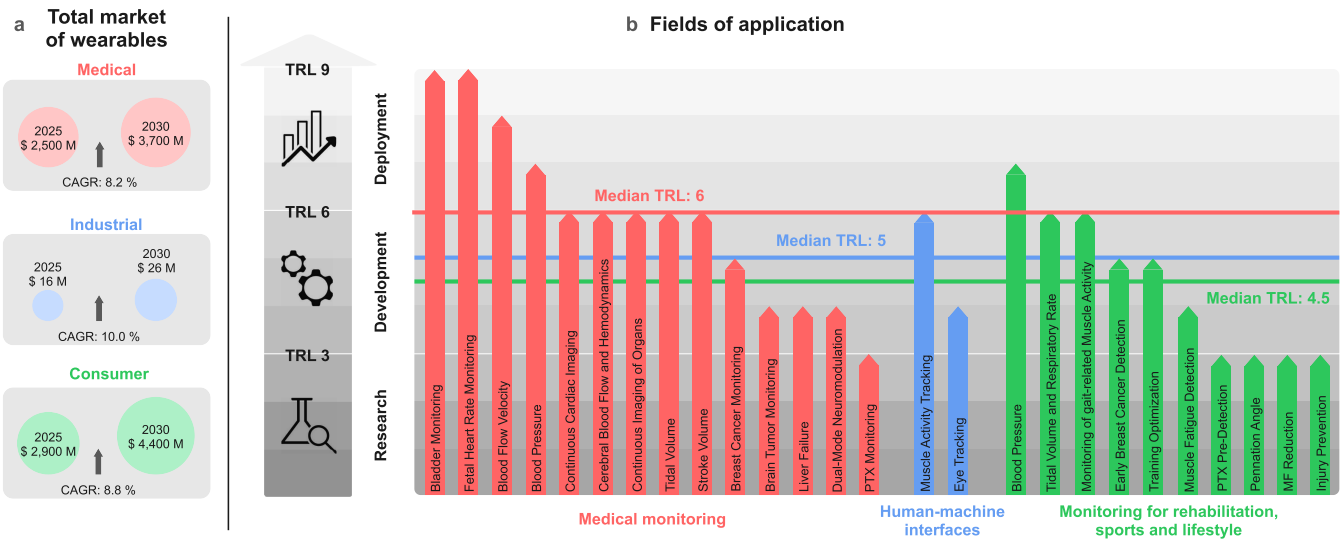


Fig. 2. Total wearable market and application fields of wearable ultrasound technologies. (a), total wearable market volume and expected growth [44], including the contribution of ultrasound wearables, subdivided into medical, industrial and consumer markets. (b), Current status of ultrasound wearable applications by technology readiness level (TRL), grouped by application domains from a system-architecture perspective. The classification scheme differs from that in a: While market data was typically reported for medical, industrial, and consumer domains, system-level considerations were better reflected by medical monitoring, human-machine interfaces, and rehabilitation, sports, and lifestyle applications. Additionally for each domain, the median TRL is depicted as a horizontal line in the corresponding color.

is already monitored for example through pulse oximetry, ECG-based impedance measurements, or simple manual counting. For both, respiratory rate and tidal volume, [15] demonstrated a wearable ultrasound-based system with a correlation greater than 99% between diaphragmatic displacement and tidal volume. Although the system of [15] is highly advanced and the application has been demonstrated on human subjects, deployment in an operational environment is still pending. The application-specific TRL is therefore classified as level 6.

To enable direct assessment of additional cardiovascular parameters, [49] developed a soft ultrasound patch for *continuous cardiac imaging*, providing real-time monitoring of stroke volume, cardiac output, and ejection fraction. It qualifies as TRL 6, as its functionality has been demonstrated on multiple human subjects, while clinical validation is still pending. So far, the system relies on a wired connection to larger external devices. Towards deployment, Sonus Microsystems and Pulsify Medical are both developing cardiac ultrasound patches, with the latter already demonstrating successful imaging [50], [51]. At the time of writing this review, neither device is commercially available.

In prenatal care, *continuous fetal heart rate monitoring* supports early detection of hypoxic events and thereby aiding timely intervention, particularly in high-risk pregnancies. For this purpose, GE HealthCare introduced the fully portable wireless system Novii+ [52], which reaches TRL 9 as it is already commercially available.

In *postoperative tumor management*, early detection of recurrence is crucial [53]. Ultrasound enables more continuous, radiation-free monitoring compared to intermittent ionizing imaging modalities. Ref. [54] developed a system that enables long-term monitoring of brain tumor growth after surgery by integrating ultrasound technology into a cranial implant. The

TRL is assessed as level 4, as validation was limited to phantoms approximating relevant tissue properties. From a hardware perspective, the system so far remains wire-connected to external devices and is not fully wearable.

Continuous monitoring of *cerebral blood flow and hemodynamics* is crucial for detecting early cerebrovascular dysfunctions and vascular pathologies, while also enabling the assessment of neurovascular functions related to sensory, motor, and cognitive processes [29]. After the early proof-of-concept study by [55], [56] demonstrated a conformable ultrasound path for stable long-term coupling and volumetric transcranial Doppler imaging towards real-time assessment of intracranial hemodynamics. Given the demonstrated functionality of the patch in multiple human subjects, but pending validation in routine clinical use, the TRL is assessed as Level 6.

A further application presented by [57] integrates ultrasound imaging and low-intensity focused *neuromodulation* by employing two dedicated transducer arrays. Such neuromodulation approaches aim to modulate the firing rates and patterns of peripheral nerves, thereby enabling controlled activation of downstream musculature and offering therapeutic relevance for conditions such as overactive bladder, neuropathic pain, and disorders related to the vagus nerve. Although this review focusses on monitoring applications, the work of [57] is considered here due to its combined use of ultrasonic sensing and modulation. Since neuromodulation was demonstrated exclusively on an in-vitro active phantom emulating peripheral nerve and muscle responses, without validation in human subjects, the system is classified as TRL 4. The current prototype remains wire-connected to external hardware and was therefore not yet realized as a fully wearable platform.

Timely detection of *acute liver failure* is of critical clinical importance, particularly following organ transplantation or post-operative intensive care. Early identification facilitates targeted therapeutic interventions, markedly decreasing consequential damage and mortality. In this field, [58] presented a wearable ultrasound system capable of continuously measuring liver stiffness via ultrasonic shear wave elastography. As validation has been limited to animal models, the system currently qualifies as TRL 4. Moreover, the system so far relies on wired connections, necessitating a future system design for a wearable integration.

Long-term imaging of inner organs can provide clinically relevant information on postoperative recovery, internal injuries, or complications without requiring repeated hospital-based examinations. To address this need, [11] introduced a bioadhesive piezoelectric ultrasound array designed for comfortable 48-hour wear and capable of stable imaging under dynamic conditions. While the system enables continuous acquisition of high-quality images on the body, it remains tethered to an external Verasonics Vantage system for data acquisition, and all recorded images require manual evaluation by clinicians. Given that the imaging modality has been demonstrated on human subjects in realistic environments, the TRL is assessed as level 6.

For the early detection of pneumothorax (PTX), [59] developed a system capable of detecting the absence of lung sliding. Early identification and continuous monitoring of pneumothorax are clinically essential, as progression can occur rapidly and may lead to severe respiratory compromise, hemodynamic instability, and the need for immediate intervention. The approach is currently at TRL 3. Further steps towards a higher level will need a quantitative validation of the results and a wearable system integration.

Ref. [60] developed a wearable ultrasound patch designed for integration into a bra for breast cancer monitoring and early detection. Initial in vivo validation in a human subject demonstrated the feasibility. Since the technology has only been demonstrated on a single human subject rather than in a broader operational setting, the current TRL is classified as level 5. In the current system, data transmission and power supply are externally wired, showing further potential towards the system's autonomy.

For urinary bladder monitoring, the primary objective is to enhance the quality of life for individuals affected by urinary incontinence [61]. Ref. [62] and [63] recently introduced systems employing wearable ultrasound. As systems are commercially available, the TRL is at 9.

B. Human-Machine Interfaces

Wearable ultrasound technology is increasingly relevant for HMI, offering competitive gesture recognition accuracy compared to surface electromyography (sEMG) by precisely capturing and localizing *muscle movements* through morphological changes [30], [64]. In contrast to sEMG, which affected by skin impedance, electrode shifts, and electro magnetic noise, A-mode ultrasound captures muscle morphology directly, providing stable features even under varying contraction levels. Ultrasound signals enable reliable recognition of deep muscle movements,

making it particularly advantageous for gesture decoding, discrete force classification, and other fine-grained motor tasks in noisy or dynamic conditions [64]. Because its signal characteristics are inherently robust to ambient electromagnetic noise and reflect mechanical rather than electrical muscle activity, ultrasound can either serve as a standalone modality or complement sEMG by adding deformation-based information that enhances overall reliability in multimodal HMI systems. Additionally, ultrasound is also enhancing eye-tracking capabilities due to its robustness against ambient light interference [65]. Based on these advantages, ultrasound-based gesture and eye recognition systems are becoming increasingly prevalent in both consumer electronics and industrial environments. The market size for industrial wearables is expected to reach \$ 26 million by 2030, driven by a CAGR of 10% , with HMI applications playing a particularly significant role [44].

In 2024, [66] introduced an ultrasonic wristband for three-dimensional pointer control, achieving markedly higher wrist-angle prediction accuracy than sEMG. Ref. [67] demonstrated motion intent decoding from unlabeled ultrasound signals with a self-supervised deep learning method, which was validated with ten probands including one transradial amputee. To enhance robustness, [68] also introduced a multimodal wearable system integrating sEMG, inertial sensors, and A-mode ultrasound.

Focusing on energy efficiency, [69], [70] demonstrated feature extraction and three-dimensional pointer control on a fully wearable low-power ultrasound system. Expanding beyond gesture recognition, [71] incorporated grip-force estimation, and [72] improved robustness to forearm-position variations through a position-invariant algorithm.

Overall, the TRL for gesture recognition applications based on muscle activity tracking is classified as Level 6, as the applications have already been validated on human subjects, while validation in an operational environment is still pending.

Eye tracking plays a critical role in enabling user interaction within AR and VR goggles. However, optical tracking methods face substantial challenges in non-sealed headset designs due to the influence of ambient light, which is compromising the functionality [65]. To address this limitation, [65] presented a laboratory demonstrator at Facebook Reality Labs, showcasing an ultrasound-based approach for gaze estimation. In their setup, static gaze angles were simulated using a physical model eye. Given that the system was only validated in a non-wearable laboratory setup using a model eye, and neither integrated into an actual headset nor tested on human subjects, its technology readiness level (TRL) is currently assessed as level 4.

C. Monitoring for Rehabilitation, Sports and Lifestyle

Wearable devices such as smartwatches, smart rings, and fitness bands are increasingly used in daily life to monitor individual well-being through parameters such as sleep quality, heart rate variability, and blood oxygen saturation [8]. The global market for consumer devices is expected to surpass \$ 4.4 billion by 2030, with a CAGR of 8.8% [44]. Also in this field, a growing market share for ultrasonic wearables is foreseeable, driven by their ability to capture biomechanical information.

Building on this growing role in personal health monitoring, wearable systems are progressively being developed to support the early detection of diseases, thereby enabling timely diagnosis and intervention. Although these devices target physiological parameters similar to those monitored by clinically deployed medical systems, they are primarily used in preventive or lifestyle-oriented contexts rather than in clinical care. This convergence in measurement targets is contrasted by divergent design priorities: Lifestyle wearables are heavily constrained by size, weight, and usability requirements, and often aim to extract multiple physiological parameters from compact sensor modules. Such multifunctionality typically comes at the cost of reduced measurement accuracy. In clinical environments, by contrast, these constraints are less stringent, allowing the use of dedicated and validated measurement modalities that offer higher accuracy and robustness. As a result of these converging measurement targets and diverging design priorities, the boundary between lifestyle and medical applications has become increasingly difficult to draw and often depends solely on the presence of official medical certification.

Beyond health-related self-monitoring, wearables also play an expanding role in rehabilitation and sports. These systems enable real-time tracking and recording of physical activity and relevant physiological parameters, thereby facilitating data-driven training optimization. With the growing adoption of such technologies across the general population, advancements in measurement accuracy or the integration of additional physiological parameters hold the potential to exert a substantial impact on public health.

Blood pressure as a key cardiovascular parameter discussed in Section II-A is also relevant for health tracking [46]. Further relevant adoptions from medical monitoring involve the early detection of conditions such as breast cancer and pneumothorax, or the monitoring of respiratory rate and tidal volume, with the latter enabling novel insights into breathing efficiency, training load and recovery in lifestyle and professional sports. The TRL for these applications is assessed similarly to their medical counterparts, as they rely on the same underlying measurable quantities.

Ref. [73] developed an algorithm capable of estimating the pennation angle without image reconstruction. The pennation angle is a relevant parameter for estimating maximum muscle force depending on the training status. This technique is currently classified as TRL 3, as only an experimental proof of concept has been demonstrated in the laboratory. Detection of muscle activity can also enable the adaptive application of spatially distributed sequential stimulation (SDSS), which was shown by [74] to reduce muscle fatigue (MF). The authors demonstrated TRL 3 of this approach, with the potential to target further technological validations in the future. Ref. [75] presented a system for predicting ground reaction forces during isometric squats in a laboratory setting, which is assessed at TRL 5. These works demonstrate that ultrasound can be used to accurately monitor muscle activity, allowing for optimization of movement patterns and individualization of training protocols. Yet, neither of these systems is available fully wearable, as they are so far operate with wired peripheral components.

Ref. [76] developed and validated a wearable system combining ultrasound, EMG, and MMG to monitor gait-related muscle activity in stroke survivors. The system enables objective tracking of rehabilitation progress, supports early detection of compensatory movement patterns, and informs timely adjustment of individualized treatment plans. As the technology has been successfully tested on patient subjects during natural movement, it is classified as TRL 6.

Beyond training optimization for rehabilitation or sport purposes, ultrasound has been employed to assess the contractile properties of skeletal muscle for injury prevention and recovery monitoring [77]. This includes contraction thickness, duration, velocity, maintenance, and half-relaxation time. The experimental proof of concept in laboratory by [77] demonstrated a TRL 3 for this approach. Ref. [78] demonstrated muscle fatigue detection using ultrasound in a laboratory environment. With this work, the authors showcase a TRL 4 for fatigue detection, as the approach was validated only in controlled laboratory experiments without application in a relevant environment. Both systems are still connected to external devices via cables.

D. Implications

Ultrasound-based wearables show a wide range of applications throughout various fields. Although market figures specifically for ultrasound-based wearables have not yet been reliably reported, their potential is underscored by the overall growth of the wearable market, which is estimated to expand into the multi-billion-dollar range with a CAGR above 8%. This suggests a favorable market environment for novel ultrasound-based applications. However, only a limited number of commercial devices are currently available.

Our TRL assessment across different domains reveals substantial variability in the maturity of ultrasound wearable technologies. While some use cases in medical monitoring, such as bladder state assessment, fetal heart-rate monitoring, and blood-flow velocity estimation, have progressed to commercial products, many applications remain at the research stage with initial pilot studies and laboratory prototypes. Even in domains where advanced TRLs have been reached through validation in relevant environments, the underlying ultrasound systems are rarely fully wearable. Instead, they remain tethered to large, high-performance, and power-intensive backend systems such as the Verasonics Vantage platform and workstation-class computers. Ultrasound wearables in the domain of human-machine interfaces typically reside at TRL 4 to 6, supported by convincing laboratory demonstrations but limited by challenges in wearability, semantic inference, and system integration. Applications in lifestyle, rehabilitation, and professional sports also remain at early TRL 3 to 7, where proof-of-concept results exist, but validation in relevant environments, manufacturability, and market fit remain uncertain.

Concluding all three domains, we identified five common challenges that mainly hinder further TRL advancements of the proposed applications:

- System-level integration to meet requirements for wearable applications,

- Fragmented solutions and the necessity for custom-build solutions raise entry barriers,
- Regulatory and safety constraints for wearable ultrasound devices [29],
- Reliable and autonomous on-body operation,
- Manufacturability and economic profitability.

These considerations are tightly coupled to system architecture and the trade-offs involved, for instance between embedded computation, wireless communication, and energy consumption. Insights from early commercial successes in handheld ultrasound further highlight that a focused use case, strong system integration, clear regulatory positioning, and reliance on commodity compute platforms can accelerate translation; principles that will likely generalize to wearable ultrasound as well. Therefore, rather than conceptual limitations of the applications, system-level constraints are the dominant bottlenecks to higher TRLs. Power consumption, safety compliance, autonomous reliability, and manufacturability challenge all application areas alike. This shared technological burden highlights the need for a more unified and strategic approach to system design in wearable ultrasound.

The considerations based on the application review motivate the structure of the following sections. In Section III, we examine the fundamental architectural principles and system-design requirements that underpin wearable ultrasound technologies, emphasizing cross-domain commonalities that arise from the challenges described above. Section IV then reviews current system implementations, highlighting architectural strategies, integration concepts, and emerging trends. Together, these analyses identify opportunities for modularity, standardization, and platform-level innovation that may enable wearable ultrasound to progress toward higher TRLs and broader societal impact.

III. SYSTEM ARCHITECTURE

During ultrasonic scans, an emitted acoustic pressure wavefront is sent from a transducer element that propagates through the medium. Variations in material properties, such as tissue boundaries or internal organ structures, cause partial reflections or backscattering, which are individually captured by each receiving element, see Fig. 3. The purpose of ultrasonic systems is to drive the transducers to perform these scans, record their signals and evaluate and transmit the relevant information. Identifying key design specifications for wearable ultrasound systems, requires an understanding of common ultrasound measurement techniques and their technological implications, which are introduced below. Building on this foundation we establish a generic system architecture whose building blocks are described in detail in the following subsections.

We propose a classification scheme for wearable ultrasound applications, dividing them into two broad categories: non-imaging and imaging systems. Non-imaging systems commonly rely on pulse-echo scanning with typically up to eight receiving (RX) channels (see Fig. 3(c)). In many designs, the connected transducers are electronically multiplexed so that a single RX channel can scan all connected transducer elements sequentially. This approach minimizes power consumption and hardware

size, making such systems particularly well-suited for wearable applications, albeit at the cost of reduced temporal resolution. Envelope detection and log-compression of the received signals produce amplitude mode (A-mode) profiles, depicting reflectivity over time or depth (see Fig. 3(e)). Building on A-mode data, repeatedly acquiring signals from a single receive channel over time produces a motion mode (M-mode) representation, which enables visualization and analysis of motion within the scanned object [79]. Similarly, a set of coherent echo profiles allows to measure flow velocities along the direction of the ultrasonic beam by evaluating the Doppler shift [79]. To achieve a higher level of information retrieval, see Fig. 3(f), further advanced methods of processing involve phase evaluation [15], spectroscopy [38], feature extraction [80] and neural network processing [69].

Imaging systems additionally involve the crucial processing part of image reconstruction (see Fig. 3(d)). This means, that they capture information coherently along at least two spatial dimensions, in contrast to non-imaging systems which only provide a single spatial dimension for each transducer element. Therefore, imaging systems typically employ 32 or more channels. In the common line-by-line (sequential) scanning approach, the transmit and receive apertures are stepped across the transducer array to acquire echo data in both the axial and lateral dimensions [79]. This method offers computational efficiency while maintaining moderate resolution. However, its sequential nature limits the achievable frame rate, and its field of view is restricted to regions near the insonation axis, which has motivated the development of plane-wave and diverging-wave scanning methods [81], [82]. Imaging systems are able to provide the same modalities as non-imaging systems (A-Mode, M-Mode, Doppler) but can also extend them to a lateral dimension with B-Modes or color Doppler images [79]. Specific system designs also enable elastography imaging [58].

Although both the non-imaging and imaging modalities described above differ in terms of their transducer configuration and achievable information content, they all rely on the same fundamental system architecture. Regardless of topology, the hardware must generate ultrasonic waves, capture and digitize the returning echoes, and process them into meaningful outputs such as depth profiles, motion traces, Doppler-derived flow velocities, frequency spectra, or images. This shared generic architecture is illustrated in Fig. 4, outlining the main functional blocks common to most ultrasound systems which include:

- A transmitting path that generates the excitation signal to be transmitted by the ultrasonic transducer,
- A receiving path that conditions and digitizes the backscattered echos,
- A central unit controlling these blocks to execute specific sequences and maintain timing constraints for the respective measurement task, and
- An integrated computing unit that either pre-processes the raw received signals before transmission to a host system or directly computes analytical results, e.g. flow velocities.

In this regard, imaging systems are merely up-scaled versions of non-imaging systems to provide the increasing amount of

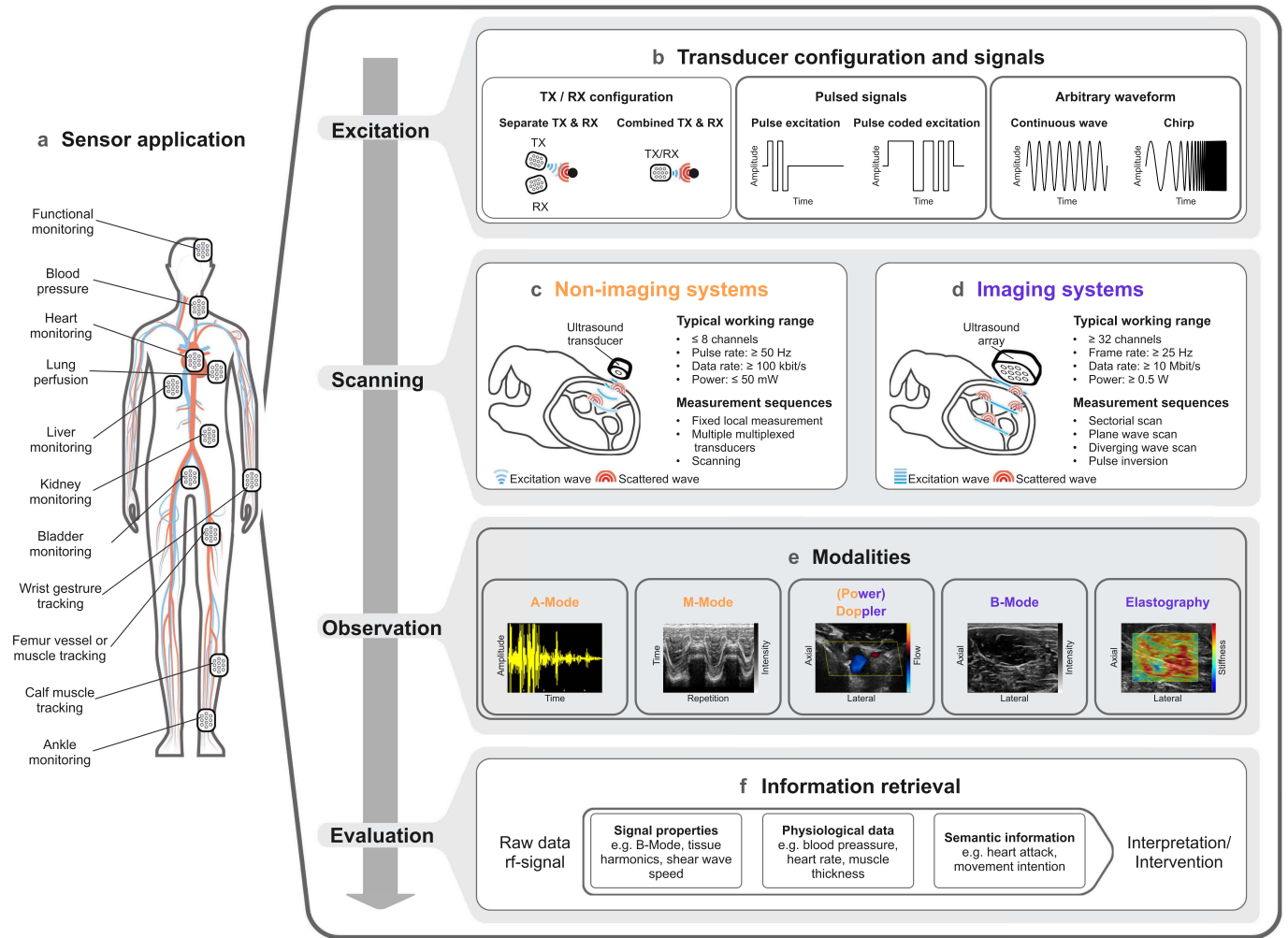


Fig. 3. Overview of the functional and methodological variety of wearable ultrasound sensor systems. (a), application variety of sensors and positions. (b), configuration transmitters (Tx), receivers (Rx) and signals of ultrasound sensor excitation. Functionality and specification of, (c), non-imaging systems and, (d), imaging systems for scanning within the human body. (e), typical imaging modalities like (A for Amplitude, M for Motion or B for Brightness Mode) acquired on which, (f), information retrieval and processing from radiofrequency (rf)-signal to interpretation is based upon. Information level in wearable systems range from raw data over acoustic signal properties and physiological data up to highly processed semantic information.

transmit (TX) and RX channels, processing capabilities, and data bandwidth (see Fig. 3(c) and (d)). The following sections describe each building block and its specific functions in more detail.

A. TX-Subsystem

Ultrasonic transducers generate acoustic pressure waves when electrically excited (Fig. 4, yellow blocks). Depending on the application, the excitation signals can be pulse trains, encoded signals, or continuous waves (see Fig. 3(b)). As depicted in Fig. 3(c) and (d), non-imaging systems can operate with just a single channel, whereas imaging systems require the coordinated excitation of multiple transducer channels. Precise control of the delay between pulses enables the generation of focused, diverging [82] or tilted plane waves beams [81], which form the basis for various beamforming and imaging schemes. The associated timing constraints can either be met by utilizing

specialized integrated circuits (ICs) [83] or a central measurement control unit, which will be described in more detail in Section III-D.

Driving ultrasound transducers also requires amplifying the excitation signal to the necessary AC voltage levels and adding, where applicable, a DC bias. In wearable systems, generating high voltage amplitudes poses significant challenges for size, power consumption, and crosstalk to the sensitive low-voltage RX circuitry. Moreover, multichannel operation can be achieved either through dedicated TX channels or by dynamically routing a single source to multiple transducers via high-voltage switches (e.g. high-voltage multiplexer (MUX)).

Beyond timing and driving electronics, the achievable performance also depends on the physical characteristics of the transducers themselves. The bandwidth, the radiation pattern and amplitude response of the excitation signal are dictated both by application needs and by the sensor employed.

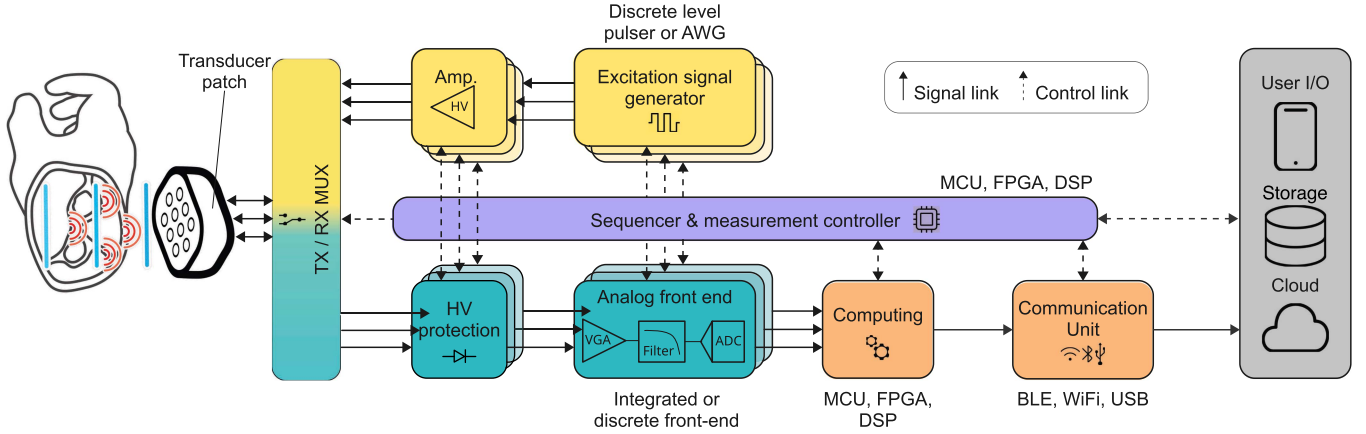


Fig. 4. Annotated block diagram illustrating the functional building blocks commonly found in ultrasound research platforms and commercial devices. The functional groups are highlighted as follows: transmit path (TX, yellow), receive path (RX, aqua), central control unit (purple), and computing unit (orange). The grey box on the right hand side illustrates peripherals that are in contact with the wearable system such as host PCs, mobile phones, tablets or cloud services. Abbreviations: ADC, analog to digital converter; AWG, arbitrary waveform generator; BLE, Bluetooth Low Energy; DSP, digital signal processor; FPGA, field-programmable gate array; HV, high voltage; MCU, microcontroller unit; MUX, multiplexer; VGA, variable gain amplifier.

In terms of operating principles, commonly used ultrasonic transducers can be broadly classified into two main categories: piezoelectric and capacitive. Piezoelectric transducers, typically fabricated from bulk ceramics such as lead zirconate titanate (PZT), are widely used due to their strong electromechanical coupling and reliable excitation performance. Conventional piezoelectric ceramics generally achieve fractional bandwidths of up to $\approx 60\%$, while single-crystal piezoelectric materials can extend this range to $\approx 80\%$ [84]. Capacitive transducers are typically realised as capacitive micro-machined ultrasonic transducer (CMUT). CMUTs, in contrast, rely on electrostatic actuation of micromachined membranes and are characterised by inherently wide bandwidths, often exceeding 100% [84], [85]. Their fabrication with standard semiconductor processes allows for close integration with digital electronics, which makes them attractive for highly miniaturised or array-based applications. However, CMUTs typically require a DC bias voltage, sometimes greater than 100 V, to operate effectively [85], [86], which can pose challenges in battery-powered or fully integrated systems.

B. RX-Subsystem

Upon reception, ultrasound transducers function in reverse: incoming echos mechanically excite the transducer elements, which convert them into weak electrical signals in the mV-range. These signals are routed through a MUX, which maps the individual transducer channels to the limited number of available analog frontend (AFE) subsystems and analog-digital-converter (ADC) channels (Fig. 4, aqua blocks). A high-voltage protection stage, typically implemented as a TX/RX switch, enables the same transducer to alternate between transmission and reception without damaging the receive electronics. After channel mapping and isolation, the signals enter the AFE, where amplification and filtering are applied to match the amplitude, dynamic range and bandwidth of the ADC, thereby maximizing sensitivity.

Direct sampling of RF data requires high-speed ADCs and a computing subsystem capable of sustaining the resulting data rates. In contrast, analog bandwidth reduction techniques [87] or band-limited excitation with down-mixing [38] can substantially lower processing requirements, enabling simpler and more efficient system designs, particularly for wearable applications [88].

C. Computing & Communication Unit

Following the digitization of the received ultrasound signals, additional computation is required to extract relevant information with following communication to the host interface (Fig. 4, orange blocks). The computation process can be divided into three subcategories: front-, mid- and back-end processing. Front-end processing operates on the raw ultrasound data and contains tasks like structuring multichannel data for further processing, reducing data-rate by applying IQ demodulation (if not already performed in the AFE) or applying beamforming algorithms. Mid-end processing builds on this by deriving quantitative physiological data or structural parameters of the specimen, such as motion, strain, or tissue properties. Back-end processing then interprets these parameters within application specific context, generating semantic information, integrating with other data sources and preparing results for visualization, transfer or storage. The communication unit establishes and maintains a secure connection to the host interface and transmits the processed data, typically using standardized protocols.

The primary objective of processing is to increase the information depth before transmitting the data to the host, as shown in Fig. 3(f). Starting from RF signals with high sampling rates and correspondingly large data volumes, computation aims to extract progressively richer information while reducing data, ultimately reaching semantic representations. For example, while ADCs in the RX subsystem with sampling rates up to 10 times the emission center frequency [79] generate output data rates of typically hundreds of Mbit/s, blood-flow velocity at a specific arterial location can be expressed with only a few bytes, and its

semantic interpretation (i.e., healthy vs. pathological) may be represented by a single bit.

This progressive reduction highlights the trade-off between local processing on the wearable system and the amount of data that must be transmitted wirelessly. The balance is highly application specific and subject to optimization. Key considerations for system design include:

- Latency due to a data link with limited bandwidth or processing with restricted on-edge resources,
- Power consumption either by a wireless interface or a processing unit, and
- Hardware cost driven by highly integrated electronics.

D. Measurement Control Unit

The sequencer and measurement control unit (Fig. 4, purple block) combines all components that orchestrate and synchronize other function blocks. The operation of this block is highly relevant to the stability and quality of the system as a whole:

- Multiple elements of TX- and RX-subsystems have to operate in tight synchronization to produce meaningful data,
- Excitation has to be synchronized with the sampling process of the ADC so that echos can be associated with a depth in the specimen, and
- Subsequent excitations must occur at a known point in time to facilitate stable frame rates as well as accurate motion and flow velocities.

The realization of this element is very application specific and can be achieved through the utilization of micro-controller units (MCUs) [15], [89], field programmable gate arrays (FPGAs) [27] or application-specific integrated circuits (ASICs). Low-level hardware control is required in order to fulfill the strict timing constraints. To configure the measurement parameters, the measurement control unit has to exchange data with the host system. In addition, a direct connection to the computing unit allows for automatic adjustments of measurement parameters based on results derived from prior measurements, e.g. selection of specific transducer channels or adjustment of repetition intervals.

E. Implications and Safety Considerations for Wearable Systems

From a systemic point of view, the functional elements of a wearable ultrasound system described above do not differ from a clinical or large research system. The main difference lies in the requirements that a wearable system must meet. Wearable devices are mostly to be used without the frequent intervention of trained personnel to allow long-term measurements in non-clinical environments. This scenario poses additional constraints on the implementation of the described functions. All building blocks must be highly integrated to realize a wearable system that does not restrict movement or mobility.

The amount of processing integrated into the device depends on the application, as the complexity can range from imaging and non-imaging modality from simple spectral processing like filtering up to neural network processing for image classification

and self-assessment of measurements. As devices should be powered by small batteries while performing long-term measurements, the power efficiency of all elements is a key design aspect. Including advanced processing algorithms directly on the wearable device can improve information throughput while reducing latency and data rate requirements for wireless transmission. To enhance integration and minimize size and power consumption, the measurement control unit is often combined with the computing unit in a single programmable device. Since wearable systems typically rely on wireless links, reducing the required data rate can significantly lower power consumption and facilitate smaller form factors. However, this comes at the cost of more complex processing architectures and memory, which in turn introduces additional power and integration requirements. These challenges are compounded by the need to co-locate these resources alongside sensitive analog electronics of the RX- and TX-subsystems within a tightly constrained physical environment.

Power efficiency is also imperative to reduce the device's heat emission in order to minimize skin heating, which can limit the usability in a wearable context. Skin heating is one of the safety considerations that presents unique challenges for wearable ultrasound systems. Unlike conventional diagnostic ultrasound systems, which typically operate for brief clinical examinations (minutes), wearable systems may be deployed continuously for hours or days, significantly increasing the likelihood of thermal and other ultrasound-induced bioeffects. Additional safety considerations include electrical and mechanical device safety, as well as the biocompatibility of materials in continuous skin contact in accordance with ISO 10993-1. The bioeffect of ultrasound radiation on tissue is characterized by the mechanical index (MI) and the thermal index (TI). The MI is related to the peak pressure of the ultrasound wave, which is typically not critical for most wearable applications. However, due to the long exposure times, the TI is typically more significant. The IEC 62359 standard provides guidelines for characterizing TI and MI. In addition, IEC 60601-2-37 defines the general safety and acoustic output requirements for diagnostic ultrasound systems, which also apply to wearable devices insofar as they function as continuous-operation imaging or monitoring systems. These requirements set baseline limits on acoustic exposure and system performance. Furthermore, the American Institute of Ultrasound in Medicine (AIUM) provides a guideline for maximum scanning times in relation to TI in [90]. Ref. [29] provides a comprehensive overview of existing norms and regulations, as well as their applicability to novel wearable ultrasound applications.

IV. STATE OF THE ART: WEARABLE DESIGN AND INTEGRATION

Building on the generic overview, this section targets an in-depth review of the current state of wearable ultrasound system design, exploring answers to the following key questions:

- What design strategies have been employed to overcome the challenges in wearable ultrasound systems?
- What trade-offs are involved?

- What trends and opportunities are emerging?

Following the taxonomy established in the previous section, we distinguish non-imaging systems (Section IV-A) and imaging systems both based on the pulse-echo method (Section IV-B). In addition, there is a growing fraction of wearable systems that are not based on traditional pulse-echo, to potentially simplify their system design and reduce power consumption, which are discussed in Section IV-C.

An overview of the systems, along with their design and specifications, is provided in Table I. Within Table I, an energy performance ratio is provided for a generic comparison of the different approaches. It is defined as the data rate divided by the power consumption, expressed in Mbit/J. This ratio is commonly used in wireless communication and sustainable computing [91] and is here adopted for wearable ultrasound systems. Since channel count, frame rate, pulse repetition frequency, and processing complexity directly affect the resulting data rate, the ratio provides an integrative measure of delivered system throughput per energetic cost. Additional information on the performance metrics and the calculation is provided in the Supplement Section 2.

A. Pulse-Echo Non-Imaging Systems

1) Closed Systems: Various non-imaging systems based on pulse-echo ultrasound already reached commercial availability (CA), helped by their lower architectural complexity compared with imaging systems (cf. Section II). A prominent example is portable fetal–maternal monitoring, where compact form factors have measurably improved wearing comfort [52]. For bladder monitoring in persons with incontinence, fully wearable A-mode solutions from [36] and [62] exemplify the category. SENS-U adheres to the lower abdomen using an adhesive and coupling gel [92], employs four A-mode transducers to realize an effective field of view of 30° , and estimates filling level by tracking bladder wall positions over time. Measurements are sent via Bluetooth to a smartphone application that issues high-fill alerts. The device weighs 55 g including the battery, supports at least 36 h of continuous operation, and uses its four-element layout to self-assess placement and provide intuitive repositioning cues; specifications are summarized in Table I. Unfortunately, due to their closed system design, despite their maturity and market availability, these devices cannot serve as generic platforms for related applications.

2) Open Research Systems: Towards generic platforms, this section focuses on open research systems that provide access to raw data, offer programming interfaces, and enable modularity. In [80] introduced the wearable multichannel A-mode ultrasound system (WMAUS). This system was subsequently employed in several works of the same group targeting hand gesture tracking for human-machine interfaces [64], [93], [94], [95]. WMAUS consist of a customized wristband with eight individual ultrasound transducers, featuring a center frequency of 5 MHz and a bandwidth of 77 %. The channels are driven in time division multiplexing (TDM) and sampled after the analog front-end at 20 MS/s. The core of the WMAUS hardware is the digital signal processor (DSP) dsPIC33, which is active

as sequencer, measurement controller and computing unit. It generates control signals for the TX and RX subsystems and performs data management and processing. The DSP collects data from a buffer and transmits it to the communication IC (Ethernet or Wi-Fi). Although the Ethernet IC supports data rates of up to 30 Mbit/s, only 0.1 Mbit/s were demonstrated, which corresponds to a frame rate of 10 Hz; likely being limited by other system factors. From there, A-mode scans were transmitted to a host PC for further processing. The hardware weighs 190 g and consumes 3.5 W, which gives an energy performance ratio of 0.03 Mbit/J. Powered by a 3000 mAh battery, the system can operate continuously for 10 h. The platform was later commercialized by [101], which paved the way for further studies in this field [71], [96], [97], [98]. Furthermore, a multimodal version that incorporates four sEMG sensors alongside four ultrasound probes in a wristband was also reported [99], [100].

Ref. [89] presented a re-design of WMAUS closely integrated into a prosthesis. It is centered around the STM32F7 microcontroller instead of the previously used DSP. In contrast, this system uses a much lower ultrasound frequency (1 MHz) and sample rate (2.4 MS/s). It weighs only 85 g and achieves an increased data rate of 1 Mbit/s. Despite the increased power consumption, the energy performance increased to 0.2 Mbit/J. With a 1500 mAh battery, continuous operation for 3.5 h is possible. To date, no statements have been made regarding the availability of WMAUS and its re-design.

In [103] introduced WULPUS, a wearable ultrasound platform for carotid artery and muscle activity monitoring. It was specifically developed to enable multi-channel A-mode scanning with low power consumption. The device weighs only 9 g and measures $46 \times 25 \times 13 \text{ mm}^3$. WULPUS is centered around the MSP430FR5043 (Texas Instruments), an ultra-low-power system-on-chip (SoC). This SoC integrates a programmable ultrasound pulse generator, an analog front-end (AFE) with amplifier and ADC, and a 16 MHz RISC CPU. The ADC samples data with 12 bit at 8 MHz. Single, trill, and multi-tone pulses can be generated with a mid-frequency of up to 4 MHz. Regarding power consumption, the SoC provides three modes: active (0.12 mA/MHz), standby (450 nA), and shutdown (30 nA). The high level of integration and low-power operation make the SoC highly attractive for wearable A-mode ultrasound applications. For this purpose, [103] complemented the SoC with a high-voltage PCB for pulse amplification, HV protection, and multiplexing to eight individual transducers. An additional SoC (nRF52832, Nordic Semiconductors) transmits the data via Bluetooth to a host PC with a data rate of 300 kbit/s. WULPUS is fully documented and available open source [106], including hardware schematics, PCB layouts, firmware, front-end and back-end software, and a detailed manual.

Refs. [69], [70] integrated WULPUS into a wearable armband with three ultrasound probes, each containing a 32-element array at 2.25 MHz. The four left- and rightmost elements of each probe were connected in parallel, resulting in six individual A-mode transducers that acquired signals at a pulse repetition rate of 25 Hz. WULPUS also demonstrates a high degree of edge processing: using embedded AI for feature extraction and

TABLE I

OVERVIEW OF STATE-OF-THE-ART SYSTEMS. AS WITH THE MAIN TEXT, THE TABLE IS DIVIDED IN THE THREE PARTS OF NON-IMAGING, IMAGING AND NON-PULSE-ECHO SYSTEMS, WHEREAS FOR EACH CONTRIBUTION THE TRANSDUCER INTERFACE, TX- AND RX-PATH, CORE HARDWARE DESIGN, THE DATA LINK AND BASIC PROPERTIES ARE SPECIFIED. AN EFFICIENCY METRIC IS CALCULATED BASED ON THE ACTUAL DATA LINK BANDWIDTH NORMALIZED TO THE POWER CONSUMPTION. FURTHERMORE, INFORMATION IS GIVEN UPON ITS DEMONSTRATED APPLICATION AND ACCESS TO THE HARDWARE.

Platform	Transducer	TX-path ¹	RX-path ¹	Design ¹	Data link ²	Specifications ²	Perf. Mbit/J	Application	Access ^{1,2}
<i>Pulse-echo non-imaging systems</i>									
<i>SENS-U</i> [36]	4 ch., integrated	n/a	4-to-1 MUX	n/a	⌘ raw data	30° FOV, 55 g, 36 h	n/a	bladder monitoring [36, 93]	CA
<i>WMAUS</i> [81]	8 ch. in wristband	±15 V, 5 MHz	8-to-1 MUX, 40 MS/s	dspIC33 (DSP) seq. and comp.	⌘ / ⌘ raw data 0.1 Mbit/s	10 Hz, 190 g, 132 × 90 × 30 mm ³ , 3.5 W, 10 h	0.03	HMI [64, 71, 94–99], sEMG hybrid [100, 101]	CA [102]; API
Yin et al. [90]	4 ch. in wristband	±50 V, 1 MHz	4-to-1 MUX, 2.4 MS/s	STM32F7 (MCU) seq. and comp.	⌘ / ⌘ raw data 1 Mbit/s	85 g, 5 W, 3.5 h	0.2	real-time prosthesis control [90, 103]	n/a
<i>WULPUS</i> [104]	8 ch. in wristband [69, 70]	15 V, up to 4 MHz	8-to-1 MUX, 8 MS/s	MSP430FR5043 (SoC) seq. and comp.	⌘ gestures and raw data 300 kbit/s	50 Hz, 13 g, 46 × 25 × 13 mm ³ , 28 mW, 3 d	11	HMI [69, 70], heart monitoring [105], EMG fusion [106]	</> [107]
<i>PuLsE</i> [89]	1 ch. in wristband	15 V, up to 10 MHz	analog envelope compression, 4 MS/s	STM32L496 (MCU) seq. and comp., analog rectifier and filter	⌘ heart rate 300 kbit/s	25 Hz, 5.8 mW, 7 d, latency 71 ms, 15 g, 4 diameter 40 mm	52	heart rate monitoring [89]	</> planned
<i>Pulse-echo imaging systems</i>									
<i>MoUsE</i> [108]	32 ch. array	up to ±100 V, 0.01 to 10 MHz, bf.	32 ch., 50 MS/s	ZYN7 (FPGA) seq. and comp.	⌘ raw data 500 Mbit/s	23 Hz, 610 g, 184 × 123 × 33 mm ³ , 12 W	40	bladder monitoring [108], wrist and hand tracking [66], prosthetic control [109]	n/a
<i>OEM USB Probe</i> [27]	128 ch. array, integrated	±80 V, up to 10 MHz, bf.	128-to-64 MUX, 40 MS/s, bf.	FPGA seq. and comp.	⌘ beam-formed data 1 Gbit/s	1 kHz, 160 × 80 × 40 mm ³ , 4 W	250	abdomen, heart, vascular, muscles, breast [27]	CA, API
<i>USoP</i> [15]	32 ch. array	up to 100 V, 6 MHz	32-to-1 MUX, 12 MS/s	PIC32 (MCU) comp., MCU seq., flexible PCB	⌘ raw data 3.4 Mbit/s	31 Hz, 100 × 30 × 5 mm ³ , 0.614 W, 12 h	5.5	mobile blood pressure monitoring [15]	schematics overview
<i>TinyProbe</i> [84]	32 ch. array	±32 V, up to 15 MHz, bf.	32 ch., 30 MS/s	STM32F4 (MCU) control and comp., Igloo2 (FPGA) seq. and comp.	⌘ raw data 21.6 Mbit/s	33 Hz, 57 × 35 × 20 mm ³ , 35 g, 0.97 W, 2 h	22	blood flow, imaging of muscles and bladder [84]	</> [110]
<i>Non-pulse-echo systems</i>									
<i>Flopatch</i> [35]	2 ch., integrated, 60° angle	continuous wave, 1 ch., 4 MHz	1 ch.	n/a	⌘ raw data	≈23 mm width FOV, 22 g, 54 × 35 × 18 mm ³ [111], 180 min [112]	n/a	blood flow monitoring in carotid artery [112]	CA
Bashatah et al. [38]	4 ch. in band	sweep 10 ms, 0.5 V, 3 to 5 MHz	4 ch. demodulator, 40 kS/s	TMS320F2 (MCU) seq. and comp.	⌘ A-mode 0.92 Mbit/s	50 Hz, 100 × 200 × 20 mm ³ , 0.82 W	1.1	muscle activity monitoring [38], ground reaction force [76]	n/a
Wang et al. [113]	11 ch. array, 2 MHz	1 ch. Barker code 5 bit, ±10 V	11-to-1 MUX, 40 MS/s	Arduino Due (MCU) seq., PC comp.	⌘ raw data	240 × 200 × 20 mm ³	n/a	bladder monitoring	n/a

¹ Abbreviations used: API *application programming interface*, bf. *beamforming*, CA *commercially available*, comp. *computation*, DSP *digital signal processor*, FPGA *field-programmable gate-array*, MCU *microcontroller unit*, MUX *multiplexer*, n/a *not available*, PCB *printed circuit board*, seq. *sequencer*, SoC *system-on-chip*.

² Symbols used: ⌘ Bluetooth™, ⌘ Ethernet, ⌘ USB™, ⌘ WiFi, frame rate, weight, form factor in length × width × height, average power consumption, battery lifetime for continuous operation, </> open source.

³ Excluding battery.

⁴ Estimation from the image provided.

classification on a separate microcontroller, the system recognizes gestures from raw ultrasound data in real time. Ref. [70] report a total power consumption of 28 mW for from ultrasound scanning to feature detection and transmission. This enables more than three days of continuous operation with a 320 mAh battery. With its lightweight design and strong energy performance of 11 Mbit/J, the system achieves gesture recognition accuracy comparable to other works that rely on larger devices consuming ≈ 70 times more power [99]. Furthermore, WULPUS has been used for heart rate and respiration monitoring [104], where the same ultrasound array was configured to combine the inner eight elements for A-mode and M-mode acquisitions.

Very recently, [88] proposed PuLSE, a compact wrist-worn ultrasound heart rate monitor. It features a single transducer channel that can be excited up to 10 MHz with a pulse repetition rate of 25 Hz. The design is straightforward: it integrates an STM32L496 MCU for digitization, sequencing, and computation, a Bluetooth module (nRF52832), an ultrasound pulser (STHVUP32), and a custom AFE [87]. With a cascade of amplifiers, diode rectifiers, and a low-pass filter, an envelope compression is performed. This lowers the necessary sampling rate to 5 MS/s, which is substantially less than what would be required for directly sampling, as the transmitted pulses have a center frequency up to 10 MHz. The MCU then directly processes the heart rate from the M-mode. This minimalistic design achieves the lowest reported power consumption in this field with only 5.8 mW, likewise achieving the highest energy performance ratio of 52 Mbit/J. The system is currently realized using the modular ultrasound development platform, Modulus [113]. This platform-based implementation enables flexible experimentation and reproducible validation of sensing and processing pipelines, while deliberately deferring form factor and weight optimization to later integration stages. In future works, integration into a watch-like device (40 mm diameter, 10 mm height, 15 g weight) is planned. Ref. [88] also announced an open-source release, which was not yet available at the time of writing this review.

B. Pulse-Echo Imaging Systems

1) Non-Wearable Origins: Research and development of wearable ultrasound imaging systems are strongly linked to the introduction of point-of-care ultrasound imaging systems (POCUS). In a POCUS device, the transducer array is integrated with all system components mentioned in Fig. 4 and connected via USB or Wi-Fi to a host device such as a smartphone or tablet [25], [26], [27].

The increasing availability of POCUS devices in recent years has accelerated the development of wearable applications and systems: [114] used a portable ultrasound device to measure muscle thickness during a treatment exercise. Ref. [115] used a POCUS device to generate biofeedback of muscle motion to improve rehabilitation outcomes. Furthermore, a POCUS device was used by [116] to provide user-specific assistance for exosuits, resulting in measurable metabolic benefits. Ref. [107] proposed MoUsE, an open platform for POCUS applications.

MoUsE has been employed by [66] as a human-machine interface for wrist and hand tracking [66] and for prosthetic control [108].

An open programmable POCUS platform, the *OEM USB Probe*, is provided by [27] for research and development of new applications, see Table I. It supports 128-channel probes with mid-frequencies up to 8.5 MHz and is fully programmable. Several probe types are available targeting medical imaging applications such as liver, brain, heart, vascular system, or muscles. In addition, it can operate 32-channel probes with mid-frequencies up to 3 MHz.

While POCUS devices are valuable for proof-of-concept studies, they are usually not suitable for wearable applications due to their size and weight. Additionally, their power consumption of typically more than 1 W requires large batteries for multi-hour use and leads to heat emission reducing wearing comfort.

2) Wearable Systems: From the pioneering POCUS devices, platforms emerged in recent years to fully target wearable ultrasound. Ref. [15] proposed USoP in [15] as one of the first wearable imaging platforms. The hardware was built around a PIC32 microcontroller (*Microchip Technology Inc.*), that served as a data acquisition module. It samples the data of one channel at once with 12 bit at 12 MHz and transmits it via a Wi-Fi IC with 3.4 Mbit/s to a host device for further processing. A second microcontroller performs the sequencing and measurement control. Only one RX channel can be recorded at once, but fast multiplexing over 32 channels allows a B-mode reconstruction on the host device. The pulse repetition frequency of USoP is 1 kHz, resulting in an overall frame rate of 31 Hz. A unique feature of USoP lies in its level of integration as a wearable device. With its flexible PCB and board connectors, it was especially developed to be worn on skin. Larger integrated components are placed on opposing sides of the PCB to increase flexibility, allowing for a maximal bending angle of 24.1° . The authors measured a power consumption of 614 mW, lasting for up to 12 h operation when powered with a “standard 3.7 V commercial lithium-polymer battery”. The energy performance of USoP accounts to 5.5 Mbit/J. A rough overview of the schematics and the PCB layout is provided in the supplements of the article [15], however, replicating or building upon the USoP design is hindered by the lack of publicly available hard- and software sources.

Recently in [83] proposed TinyProbe. It is a highly compact imaging platform with dimensions of $57 \times 35 \times 20 \text{ mm}^3$ and a power consumption range of 0.4 to 1.2 W. The actual power consumption depends on the mode of operation and the used pulse repetition frequency. For a typical imaging sequence, TinyProbe consumes 0.97 W with an energy performance ratio of 22 Mbit/J, for which [83] ensure a 2 h battery lifetime. At its core, a microcontroller of the STM32F4 family controls the Igloo2 FPGA sequencer and buffers the data for the Wi-Fi IC to be transmitted to a host. So far, the authors only used TinyProbe for raw data acquisition with offline post processing on a host PC. Nevertheless, an SoC for edge-computing applications [117] is already integrated, which the authors will evaluate in future works. It is also worth mentioning that TinyProbe is the first

platform of that size, weight and power consumption to incorporate 32 channel TX beamforming and RX recording. Firmware, schematics and design files of TinyProbe are available open source [109].

C. Non Pulse-Echo Systems

Medical ultrasound systems typically use high-frequency, short-duration pulses on combined TX/RX transducers to achieve spatially resolved measurements. Due to the strong attenuation of ultrasound waves in the human body, high-voltage pulses in transmit and low-noise amplifiers in the receiving path are required to achieve a sufficient signal-to-noise ratio (SNR). Furthermore, the analog to digital conversion typically must occur at least at four times the center frequency of the transmitted signal to capture the full bandwidth of the echo signal [118]. The power consumption of high-voltage pulse generators and the digitization and transfer of large amounts of data pose major challenges. This section focuses on systems and approaches that employ other signal types to address some of the shortcomings of pulse-echo ultrasound.

1) Continuous Wave Approaches: Refs. [47], [111], [119], [120] and [121] introduced an adhesive continuous wave Doppler patch for real-time monitoring of blood flow velocity in the common carotid artery. The device uses two 4 MHz transducers for continuous transmission and reception. A homodyne demodulator extracts the Doppler signal from the received RF data into an audio-band signal. Every 12.8 ms a new Doppler signal is extracted, providing near-real-time data essential for capturing dynamic blood flow changes in clinical settings. This approach is beneficial for the integration in a wearable, as the data processing and communication for audio-band signals in the kHz-range operates with much lower sampling rates than ultrasound signals in the MHz-range. Therefore, small-sized and efficient custom ASICs or FPGAs could be employed as the computation unit [122]. The system is commercially available through [35].

In context of Doppler blood flow measurement, the estimated velocity depends on the inclination angle between the ultrasound beam and the vessel direction, which is typically only known approximately. To reduce the errors of a mispredicted angle, [48] proposed a 3×3 angled transducer patch that allows for estimating the inclination angle. However, no fully wearable device was demonstrated, as it relies on external equipment to operate.

2) Chirp Approaches: In [38] proposed time-delay spectrometry for low-voltage dynamic tissue imaging. A continuous linear chirp is used as excitation signal, where the excitation frequency increases over time, see Fig. 3(b). The chirp propagates into the tissue, where each scatterer returns a time-delayed, attenuated echo of the signal. The time delay, which is proportional to depth, correlates to a frequency difference. An increasing depth results in a longer delay and a higher frequency shift between the transmitted and received signals. Using demodulation and Fast Fourier Transform, an A-mode is reconstructed from the received signal. In their design, [38] employed chirps from 3 MHz to 5 MHz in 10 ms that are amplified to a peak-to-peak voltage of 5 V. The received signal

is demodulated into a 100 kHz baseband signal by a custom analog circuit. For digitization, a 40 kS/s ADC is sufficient, resulting in a three orders of magnitude lower data rate compared to typical pulse-echo systems. As a result, the MCU-integrated ADC could be used, reducing the overall system's complexity (cf. Table I). Ref. [38] demonstrated the practical feasibility by achieving simultaneous four-channel ultrasound imaging at a frame rate of 50 Hz with a power consumption of 0.82 W, corresponding with an energy performance ratio of 1.1 Mbit/J. As the results were obtained with development boards, a much lower power consumption and smaller form factor can be expected through further integration and optimization. No information on the availability of code and hardware designs is given in the paper.

Ref. [123] introduced a stepped chirp for ultrasound sensing. Targeting early implant-failure detection, their *in silico* study uses separate transmit and receive units for simultaneous TX and RX. Eight small piezo-transducers were embedded in a flexible cuff that is tightened around the joint of interest. An ultrasound low frequency chirp from 1 to 300 kHz in 1 kHz steps was used to scan the cortical bone and the implant interface. A machine learning algorithm detected and classified possible hip implant failure and its severity. With this study, [123] demonstrated the potential of power efficient low-frequency ultrasound and data-driven processing for inexpensive point-of-care screening of implant integrity. However, lacking a capable hardware platform, the study was only conducted *in silico*.

3) Coded Excitation Approaches: Coded excitation schemes typically incorporate pulse compression [124], [125], a technique originating in radar systems, to spread the same transmit energy over a longer, lower-amplitude waveform and subsequently recover axial resolution during post-processing. This makes them particularly attractive for wearable systems, where the stringent requirements for power efficiency, form factor, and cost necessitate drastic compromises, especially with respect to acoustical performance [126] and system design. The resulting SNR loss can be compensated through coding gain; however, it strongly depends on the type of signal employed and further influences the complexity of the analog electronics as well as the spatiotemporal resolution.

In [112] proposed an ultrasound system as a pre-voiding alarm system for children with nocturnal enuresis based on Barker coded pulse excitation. Barker codes are two-level (± 1) pulse-compression sequences that allow for low-voltage excitation amplitudes [125]. The authors used a 5 bit Barker code that is phase-modulated on a seven-cycle 1.95 MHz continuous sine wave and amplified to ± 10 V for transmission. A linear array consisting of eleven square ultrasound transducer elements was employed, where sequential pairwise transmit-receive operations were performed and processed through a matched filter. By means of pulse compression, an SNR gain of 12.6 dB was achieved. The prototype was implemented using development boards, a large digitization device, and computation on a PC via USB, but the approach shows potential for further integration towards reduced power consumption and smaller form factors. No statements were made regarding the availability of the proposed system.

Another Barker-code-based pulse compression method was proposed by [127]. In their *in silico* study, a transducer with a center frequency of 10 MHz was used to transmit a 13-bit Barker code modulated with a bipolar pulse. In contrast to previous works, [127] employed mismatched filtering combined with in-phase quadrature compression. The authors demonstrated that this approach enables efficient data compression with reduced computational complexity while maintaining image quality.

Ref. [128] demonstrated Golay-coded excitation for rotational intravascular ultrasound to cancel range sidelobes while preserving axial resolution using weighted averaging compression. Based on this, [129] implemented realtime ultrasound biomicroscopy at 30 MHz using 4-, 8-, and 16-bit Golay pairs, increasing detection depth and SNR without degrading axial resolution. However, a translation of Golay- or Barker-coded excitation to wearable ultrasound devices remains to be demonstrated.

Although not in a medical context, [130] presented a novel low-power hardware architecture based on long pseudo-periodic coded excitation. The approach drastically reduces electronic complexity by directly coupling transmission and receiving circuitry, operating them simultaneously within the same low-voltage range (± 2 V). This eliminates the need for high-voltage generation, transmit–receive switching, and low-noise amplification, albeit at the cost of a severely reduced SNR in the raw signals. The recorded waveforms are dominated by the transmit signal with weak superimposed echoes; however, through matched filtering the impulse response of the specimen can be reconstructed and coding gain realized. In their proof-of-concept demonstrations for non-destructive testing, the authors achieved comparable SNR to state-of-the-art high-voltage ultrasound equipment (200 V excitation) while operating at low voltage, and further showed the feasibility of multi-channel operation using quasi-orthogonal TOP-CS codes. So far, the system has only been realized with laboratory-grade equipment, and no translation to medical or wearable applications has been demonstrated. Nevertheless, the combination of ultra-low excitation voltage and simplified hardware requirements makes a translation to wearable ultrasound systems desirable.

D. Summary

This section concludes with a concise overview addressing the initial questions on design strategies, trade-offs, and emerging trends.

1) Design Strategies: Wearable ultrasound systems increasingly rely on high integration, using MCUs, DSPs, and low-power SoCs to combine pulse generation, reception, digitization, and processing into compact electronics. Flexible PCBs and distributed component placement help achieve skin-conformal form factors without sacrificing essential functionality [15]. To reduce data rates and complexity, many systems adopt strategies such as envelope detection [88], analog demodulation [38], or pulse compression [112]. Modular open source platforms [106], [109] enable raw data access, user programmability, and reconfigurable sensing setups for research applications. Edge processing [38], [88] and embedded AI [69], [70] are becoming central design elements, allowing devices to interpret signals locally and

reduce wireless transmission load to optimize energy efficiency and latency. Together, these strategies enable increasingly compact, efficient, and versatile wearable ultrasound devices.

2) Trade-Offs: Balancing functionality within ultra-low power demands and size constraints represents the key design trade-off for the ultrasound wearables investigated, which is characteristic of wearable systems in general. What is unique are the chosen design routes. Decreasing the number of RX channels and relying on multiplexing is an effective compromise [36], [80], [103], [112]. However, this drastically limits the achievable framerate of imaging systems [15]. Ref. [83] showed that the number of RX channels can be increased using dedicated low-power integrated circuits. Large data volumes necessitate fast wireless links [15], [83], [89] or on-device compression [38], [70], [88], shifting power consumption to communication or computation circuitry. Flexible and thin mechanical designs improve comfort but restrict component placement, cooling, and high-voltage isolation [15]. Fully integrated closed systems are efficient but lack configurability [35], [36], whereas open platforms are adaptable but often bulkier and less optimized [106], [109]. Overall, these competing constraints shape the architectures chosen for wearable implementations.

3) Trends & Opportunities: The field is trending toward edge-intelligent wearables that perform real-time inference on-device, reducing data transmission requirements and enabling autonomous operation [70], [88]. Publications are emerging with low-voltage and low-data-rate ultrasound approaches, such as continuous wave [119], [121], chirp scanning [38], [123], and coded excitation [112], [127], [130], which potentially simplify hardware and lower power consumption. Miniaturized multi-channel systems now incorporate low-power FPGAs and/or MCUs [15], [83], pushing imaging systems closer to practicality whilst maintaining a high degree of flexibility. Open-source hardware platforms are expanding access to high-quality imaging and sensing tools for research and innovation [88], [106], [109]. Flexible, skin-conformal electronics are enabling long-term physiological monitoring in everyday settings [15].

V. DISCUSSION

A. Current Limitations

One of the central limitations comes with the capabilities wearable systems can achieve given the strict demands in form factor and power efficiency. Due to its simplicity and data efficiency, non-imaging ultrasound modalities are highly relevant for wearable applications [46], [80], [92], [103]. Information about movements, pulsations and physiological changes can be acquired from deep inside the body. Non-imaging systems were shown to provide accurate results for HMI applications [80], [99], [108] and physiological monitoring [15], [88], [104], [131]. They can be highly integrated to a lightweight system with low-power consumption, enabling a multi-day use of a fully wearable device with only a single charge [70], [104]. As a consequence, up to now available and fully wearable devices are based on non-imaging scans [36], [52], [62]. The major disadvantage of non-imaging modalities is the uncertainty of spatial localization, requiring well-trained personnel or automated

evaluation and feedback for sensor placement [36]. This is manageable for larger objects like bladders, but is problematic for the placement on finer structures such as the heart, muscles and arteries.

Imaging systems on the other hand allow for a tracing and compensation of motion facilitating an observation and evaluation of complex physiological changes and patterns. From a transducer perspective, wearable imaging has seen significant advancements towards conformable sensors. Examples include the flat-panel PMUT array developed by [51], the BAUS-E system by [58] for wearable elastography, and transducer arrays based on liquid metal electrodes [49], advanced piezoelectric ceramics [132], lead-free piezoelectric polymers [133], or silicon CMUT arrays [134]. However, to drive these transducers a platform capable to excite, record and process these multiple channel transducer arrays is required. Wearable elastography [58] furthermore requires push-beam excitations as well as the capturing and processing of more than 1000 images per second, which is far beyond the capabilities of state of the art wearable systems [15], [83]. Further limitations are given in the data bandwidth and real-time processing capability required, which is yet hardly possible to be achieved within form factors, power consumption and heat emission that wearable applications demand. As a result, all these advanced transducers have so far been operated using large research platforms. Therefore, it is reasonable to assume that the first applications of wearable imaging transducers will emerge in clinical settings on immobile patients, where the use of a larger platform is not a limiting factor. For non-imaging transducers, clinical studies already exist [46], and for imaging applications, it is likely only a matter of a few years before similar studies are published.

The introduction of wearable imaging transducers for monitoring purposes in hospitals within the next five years therefore appears realistic, especially since, for example, Pulsify Medical's wired transducers are already approved as medical devices [50]. However, fully wearable systems intended for use outside of clinical environments on mobile patients still require considerable research and development efforts in system design. This is in contrast to the non-imaging systems, where already several products employing single transducers are available and completely wearable [36], [52], [62].

B. System-Level Considerations

Our analysis reveals that the majority of limitations, especially for more advanced applications and transducers, are limitations of the system design. Therefore in this section we further analyze recurring architectural considerations across wearable ultrasound systems.

1) Design Trade-Offs: Wearable devices necessitate a trade-off between two conflicting objectives:

- The necessity to integrate complex processing units [20], which pose challenges in integration, power supply complexity and memory requirements.
- The concern of reducing data-link bandwidth [103], particularly in wireless implementations where power consumption is a crucial factor.

This critical trade-off for wearable system design needs a complex balance of energy consumption, heat emission, data rate, computational workload and form factor [16]. On one hand, computational workload on-edge towards semantic information is beneficial for a lower data throughput, allowing for low-power communication standards as for instance Bluetooth [69]. But on the other hand, large computation workloads may increase energy consumption and heat emission above a level that is feasible for a wearable system. For imaging systems, this is even more critical because of the increased number of channels to be excited, recorded and processed. Hence, the challenges to the integration of a fully wearable imaging device are much more severe compared to non-imaging devices [83]. Two potentially wearable imaging platforms were introduced so far, namely USoP [15] and TinyProbe [83]. They can serve as open research platforms given their availability. Yet, their data acquisition, computation and link-to-host capabilities are not sufficient to exploit the potential given by recently published conformable transducer arrays. For instance, to correct all degrees of freedom of the conformable array of [49], a platform to drive all 348 elements individually including TX- and RX-beamforming is necessary. This equals eleven TinyProbe platforms, adding up to a power consumption of ≈ 11 W. Sustained operation above 1 W is generally impractical for compact, skin-worn devices with multi-hour battery life targets, because it implies large batteries and increased heat dissipation. One key bottleneck is technological: no integrated circuit currently combines an ultrasound-grade analog front-end, the low-power control and tightly coupled memory architecture typical for microcontrollers, and the high-speed, sequencing and parallel processing capabilities of FPGAs into a single IC optimized for low-power wearable-ultrasound. In the case of TinyProbe [83], this functionality is distributed across three separate ICs, namely an analog front-end, a microcontroller, and an FPGA. Integrating these functions into a dedicated wearable-ultrasound system-in-package (SIP) or ASIC would reduce inter-chip communication overhead and board-level complexity, and is therefore expected to significantly reduce energy consumption and physical dimensions.

Considering parameters such as the achievable computer-milliwatt of existing microcontroller units, the bandwidth and energy cost of commercial wireless links, and the PCB footprint of programmable front-end components indicates that modular, scalable platforms are technically attainable for low-to mid-complexity sensing tasks. For full imaging pipelines, especially conformable arrays, they are only marginally feasible with current technology. Bridging this gap will require both architectural innovation and advances in device technology, potentially complemented by alternative operating paradigms such as non-pulse-echo imaging schemes or energy-harvesting-assisted designs [135].

2) Unsupervised Reliability: Long-term unsupervised measurements require a stable measurement setup and, potentially, the self-assessment of recorded data [29]. This can be realized by either transferring ultrasound data to a host system for further evaluation or by integrating advanced processing capabilities into the wearable device [20], [69]. Specifically

for imaging systems, conformable transducer patches enabled worn, unsupervised operation in the first place, but for reliable image reconstruction accurate self-calibration of the deformable transducer elements is required [31]. So far, several self-calibration approaches were reported, such as software based shape estimation [136], optical shape sensing fibers [137] or electrical sensing of element-intercapacity-changes [134]. Each of them more or less increases system complexity and adds to the computational workload [31], increasing the dilemma concerning data-rate and on-edge processing capabilities within a limited power consumption. Overall, bringing applications into the unsupervised wearable market requires a whole new level of functional integration and stability to be operated by non-experts.

3) Computation and Semantic Abstraction: Unlike handheld ultrasound systems that focus primarily on raw signal capture or imaging [23], [25], [26], [27], wearable ultrasound often directly targets higher-level semantic information. Without a trained operator, who controls the device and interprets the results, the wearable system must directly evaluate physiological states [46], [120], health prediction [49] or user intentions [3], [69]. The corresponding processing pipeline typically spans multiple stages: from raw signal and feature processing, to the estimation of physiological quantities (e.g., tissue stiffness, blood flow velocity), and up to the inference of semantic-level descriptors (see Fig. 3).

Systems that perform only basic signal conditioning on-edge and transmit raw data for host processing provide a high level of flexibility [15], [80]. In turn, a high data rate is required which can become a bottle-neck for frame rate and scanned depth and can consume more power [83]. In this direction, this review shows that much of the potential lies in the high level of integration and smart edge processing [20]. This is exemplary shown by WULPUS [69], [70], [103]. It produces results comparable to those of WMAUS [80] but with 22 mW it consumes ≈ 70 times less power. The recently published PuLSE system [88] also goes into this direction, showcasing even less power consumption (5.8 mW) to drive a complete ultrasound heart rate monitor.

Computation and semantic abstraction can be considered as one of the key design challenges for wearable ultrasound systems. It may operate in the domains of edge, host and cloud processing, but for its key requirements of weight, power consumption, latency and runtime the main focus lies on the edge processing. Here, the level of edge processing varies from the mere output of raw data [15], [80], to beamformed data [27] and up to physiological data [88], [103]. Interestingly, the power consumption of these systems is drastically reduced towards higher semantic information processing on-edge.

4) Interfaces: All analyzed systems in this review used available communication interfaces such as USB [27], [38], [107], [112] and ethernet ICs [80], [89] for the wired systems or WiFi [15], [83] and Bluetooth ICs [35], [36], [88], [103] for the wireless systems. Here, the wireless communication interfaces are particularly relevant for wearables. Systems that employed a Bluetooth interface were able to achieve the lowest power consumption [88], [103]. Bluetooth has a considerable lower

data rate than WiFi, but the ICs consume much less power. If the system is able to scale down to the lower Bluetooth data rate, low power systems that work for several hours are achievable [69], [88]. Here, ultrasound wearables can benefit from the developments in communication standards and ICs drawn from other wearables and IoT devices. This approach enables fast prototyping, but encounters limitations for clinical environments that operate on DICOM and PACS standards.

5) Non Pulse-Echo Systems: Most works on ultrasound wearables have focused primarily on pulse-echo systems. However, approaches using alternative signal architectures allow to address the unique challenges of wearable ultrasound [47], [121] through shifting the complexity from hardware towards signal processing. Techniques such as information encoding and on-edge analog or digital signal processing can significantly reduce power consumption and minimize the number of required integrated components [38], [112], [127]. These approaches also influence the design and integration of transducers in new ways.

So far, only a few demonstrations or commercial products, such as those by [35] and [38], have explored these non pulse-echo approaches for specific applications. As such, we believe that future research in this direction can unlock further potential for compact, lightweight and efficient ultrasound wearables.

VI. CONCLUSION

Wearable ultrasound sensing can address major clinical, technological, and societal challenges, ranging from personalized healthcare and rehabilitation to industrial human-machine interaction. Market projections indicate sustained growth. However, the transition from application-specific requirements, through laboratory prototypes, to widely adopted products remains difficult. Our assessment of potential applications and their development stages revealed five major challenges that hinder the transition to market-ready products, namely integration, fragmentation, regulatory and safety constraints, reliability, and manufacturability within economic profitability. Above all, the gap is primarily defined by unresolved system-level constraints. The challenges of system design span multiple disciplines and require coordinated progress in analog/digital IC design, micro-machined transducers, low-power signal processing, edge AI, packaging and microsystems engineering, and scalable manufacturing.

Based on the bottlenecks, market insights, anticipated requirements, and the current state of the art discussed in this review, we outline a phased roadmap for the future development of wearable ultrasound systems:

**1) Short-Term Priorities (1 to 3 Years) - Foundations for Com-
patibility and Rapid Prototyping:** The development of wearable ultrasound systems is currently hindered by the fragmented nature of individual solutions. Many projects are forced to design custom electronics and system infrastructure from the ground up [1], [32]. This requirement consumes substantial time and resources and effectively restricts progress to research groups with expertise in electronic and system design, hindering interdisciplinary collaboration across engineering, scientific, and clinical disciplines required for wearable systems [16], [29].

This review showed, that wearable ultrasound systems share considerable common ground in their underlying architectures, indicating that limited availability, re-usability, and flexibility of existing hardware platforms represent a key bottleneck to near-term progress. Over the next three years, this challenge can be addressed by prioritizing shared, open, modular, and reconfigurable development platforms. These platforms should integrate front-end circuitry, embedded processing, and wireless communication to lower entry barriers for application-oriented teams and accelerate rapid prototyping, as it was demonstrated with the Modulus platform [113] to develop PuLSE [88]. Furthermore, a platform-based approach enables direct comparability of results across different research efforts and allows for early identification of system requirements and solutions. As a next step towards higher integration, generic research platforms can act as testbeds for more specialized, highly efficient ASIC implementations.

Additionally, standardization of interfaces, datasets, and benchmarking protocols will foster comparability across research groups and support more systematic design decisions. These efforts should cover signal quality, energy consumption, thermal behavior, and mechanical robustness. At the signal-processing level, preliminary semantic information processing will help reduce data bandwidth and mitigate power demands of communication units. In parallel, advances in conformable packaging will improve multi-hour stability and user comfort. Together, these efforts will help elevate current systems from early proof-of-concept stages towards TRLs 4 to 5.

2) *Mid-Term Development Goals (3 to 7 Years) - Integration, Specialization and Regulatory Alignment:* Mid-term progress will rely on deeper system integration and the emergence of application-specific architectures. Critical milestones include the development of dedicated ultrasound SIPs or ASICs that integrate transmit/receive front-ends, analog-to-digital conversion, beamforming accelerators, and low-power control logic into highly compact and energy-efficient packages. Such integration will be necessary to support scalable arrays, higher channel counts, and more advanced signal architectures. In parallel, distributed sensing concepts are expected to emerge, where multiple miniaturized ultrasound nodes operate cooperatively on the body and exchange data over human body area networks. This distributed approach enables spatially adaptive sensing, reduces per-node complexity, and allows computational load to be shared between nodes and central hubs. Tight coupling of sensing, local preprocessing, and low-power communication will be essential to maintain energy efficiency and latency constraints. Hybrid sensing approaches that combine ultrasound with inertial, electrophysiological, or optical modalities will further improve robustness and inference quality, especially in human-machine interaction and physiological monitoring. Advances in packaging and biocompatible materials will enable reliable operation over 24 to 72 hours, addressing motion-induced artifacts, moisture exposure, and long-term adhesion. Embedded semantic-processing engines based on specialized AI accelerators will allow devices to output intermediate or high-level information rather than raw signals. Concurrently, regulatory pathways for continuous ultrasound exposure must

be clarified, including thermal and acoustic safety envelopes tailored to wearable operation. Achieving these goals will move wearable ultrasound devices into TRLs 6 to 7, making them suitable for clinical and industrial pilot deployments.

3) *Long-Term Vision (7 to 10 Years) - Fully Integrated, Autonomous, and Scalable Wearable Ultrasound Systems:* In the long term, wearable ultrasound systems will evolve into fully integrated, autonomous platforms capable of continuous operation in daily life. This will require the development of ultra-low-power system-on-chip solutions that unify analog front-ends, digital processing, memory, AI accelerators, and wireless communication into a single, energy-efficient package. Large-area flexible transducer arrays, enabled by mature CMUT/PMUT-on-flex technologies and wafer-level integration methods, will conform seamlessly to the body and enable new form factors, including multipatch distributed sensing. Such systems may collaborate wirelessly to achieve synthetic-aperture or distributed-beamforming capabilities, enabling imaging or semantic extraction at the system level rather than at a single device. Autonomous on-body semantic inference pipelines will allow devices to directly output physiological states, motion intentions, or early warning indicators, minimizing data transmission and enhancing privacy. These advances will open the door to personalized medicine and rehabilitation routines, and human-robot collaboration. Ultimately, this phase corresponds to TRL 8 to 9, marking the arrival of mature, manufacturable, and widely deployable wearable ultrasound technologies.

Together, this roadmap shows the critical technological, interdisciplinary and infrastructural steps required to transition wearable ultrasound from isolated prototypes and proof-of-concept demonstrations to scalable, reliable and impactful systems across healthcare, industry, and consumer domains.

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